Introduction

System identification

The exact Hankel singular-value spectrum, obtained via same-seed impulse subtraction (a theoretical reference not physically realisable with a real brain), has rank 6 (Figure 1). A physically realisable single-run OLS estimate ($N = 5{,}000$ steps) raises a noise floor that obscures modes 3–6 (exact values $\sigma_3/\sigma_1 = 0.125$ down to $\sigma_6/\sigma_1 = 3.3 \times 10^{-3}$); BIC model selection recovers only $n_x = 2$. These modes carry little input–output energy and are near-unobservable from any finite honest recording. However, the downstream decoder operates on frequency-selective latent activity, so a model that misrepresents the transfer function will produce incorrect muscle activations regardless of one-step prediction quality. ERA+EM was therefore fitted at $n_x \in \{2, 4, 6\}$ to identify the minimum sufficient order.

At $n_x = 2$ the model reproduces per-neuron frequency selectivity for u_1 but not for u_2 (Figure 5). At $n_x = 4$ both input channels are recovered; $n_x = 6$ yields no visible further improvement. ERA+EM at $n_x = 4$ was therefore selected. The Bode magnitude error (Figure 3), computed diagnostically via same-seed impulse subtraction, confirms this pattern: mean error falls substantially from $n_x = 2$ to $n_x = 4$, with negligible reduction at $n_x = 6$.

CVA+EM was fitted at $n_x = 4$ for comparison. Its transfer function reproduces the u_1 selectivity but misrepresents u_2 relative to the honest OLS estimate (rightmost column, Figure 5). One-step-ahead Kalman prediction on $N = 100$ held-out trials ($T = 2{,}000$ steps each) confirmed ERA+EM superiority across all output channels (Figure 4). ERA+EM at $n_x = 4$ was adopted; the identified matrices are shown in Figure 2.

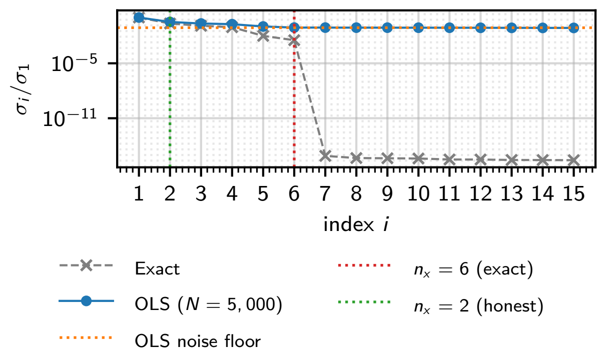

Figure 1: Normalised Hankel singular values σ_i/σ_1 for the 40×60 block Hankel matrix ($n_u = 2$, $n_y = 16$). The exact spectrum is obtained by same-seed impulse subtraction, which is not physically realisable, and serves as a theoretical reference for the true system order. Modes 3–6 fall below the OLS noise floor of a 5000-step physically realisable recording; BIC model selection on the OLS estimates recovers $n_x = 2$. Source

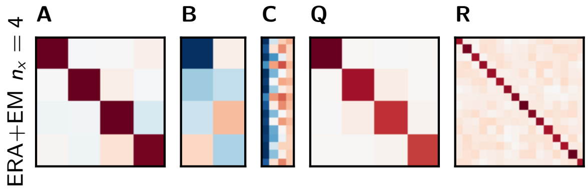

Figure 2: Identified system matrices for ERA+EM ($n_x = 4$). Red indicates positive, blue negative; colour scales are symmetric about zero. Source

Closed-loop spectral control

Having selected ERA+EM at $n_x = 4$ as the working model, we test whether closed-loop stimulation can reproduce a prescribed frequency spectrum in neural population activity. The target is a sum of two sinusoids at $f \in \{0.01, 0.03\}$ cycles per sample with amplitudes $\{1.5, 0.7\}$, oscillating about a physical population mean of 2.5 (well within the reachable range of $[0.24, 5.4]$). Three controllers were implemented against this model: Linear–Quadratic–Gaussian (LQG), LQG with integral action (LQI), and Model Predictive Control (MPC).

All three use a Kalman filter for online state estimation. The LQR tracking cost is a per-neuron cost $C^\top C$ ($\lambda_{\text{var}} = 1$), driving all 16 output channels simultaneously. LQI adds an integrator on the centred output error to reject steady-state bias; the integrator weight q_{int} is kept small so that sinusoidal tracking error is not accumulated. MPC solves a condensed receding-horizon quadratic programme ($H = 30$ steps, covering one full period of $f = 0.03$) with explicit input box constraints, warm-started between successive steps. Controller hyperparameters were selected via a log-uniform random search (8 trials per controller) scored on RMS tracking error.

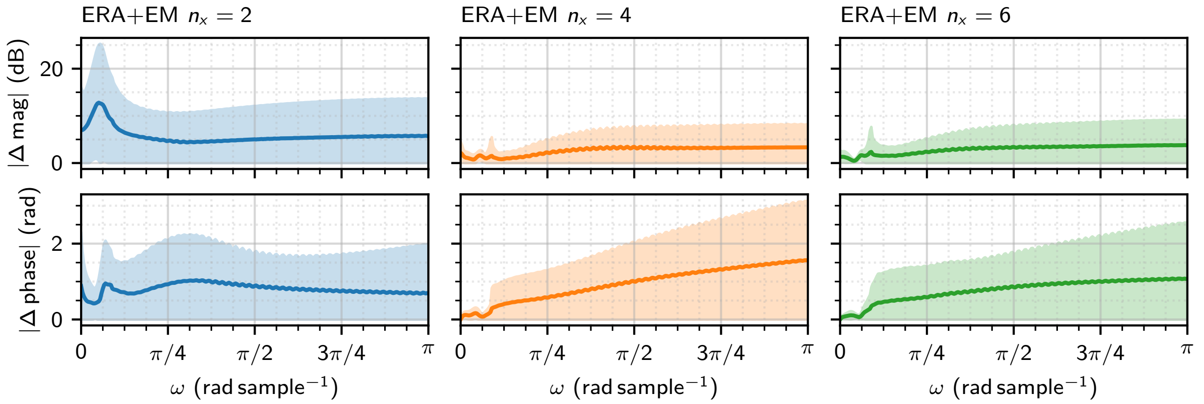

Figure 3: Magnitude error (top row) and phase error (bottom row) of ERA+EM transfer functions for $n_x \in \{2, 4, 6\}$ (left to right). Each panel shows the absolute error between the exact empirical transfer function and the analytic model response $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$, as mean $\pm$ 1 s.d. across all $n_u n_y = 32$ output–input channels. The empirical reference is the exact discrete-time frequency response obtained via impulse subtraction (same-seed noise cancellation), DFT zero-padded to 512 points. Columns share a common x-axis; rows share a common y-axis. Source

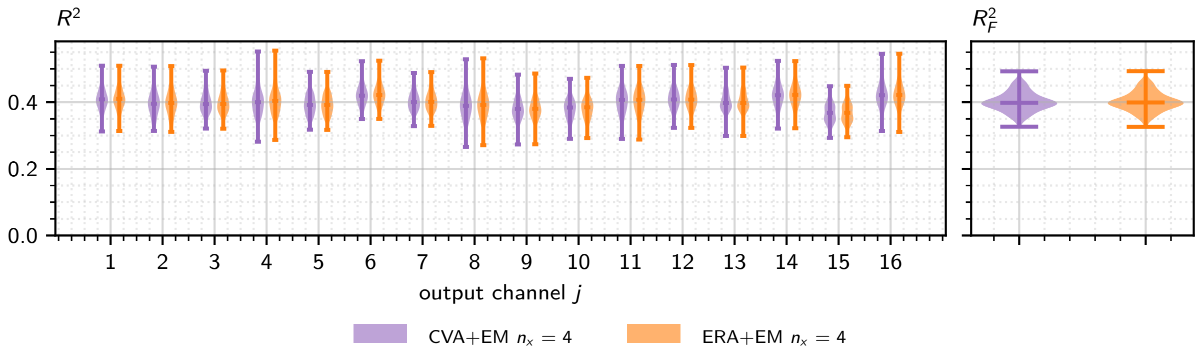

Figure 4: Per-channel R^2 (left, $n_y = 16$) and Frobenius R_F^2 (right) of one-step-ahead Kalman predictions for CVA+EM and ERA+EM at $n_x = 4$, across $N = 100$ held-out trials of $T = 2000$ steps (first 100 steps excluded as Kalman transient). Median: CVA+EM $n_x = 4$ $R_F^2 = 0.398$; ERA+EM $n_x = 4$ $R_F^2 = 0.399$. Source

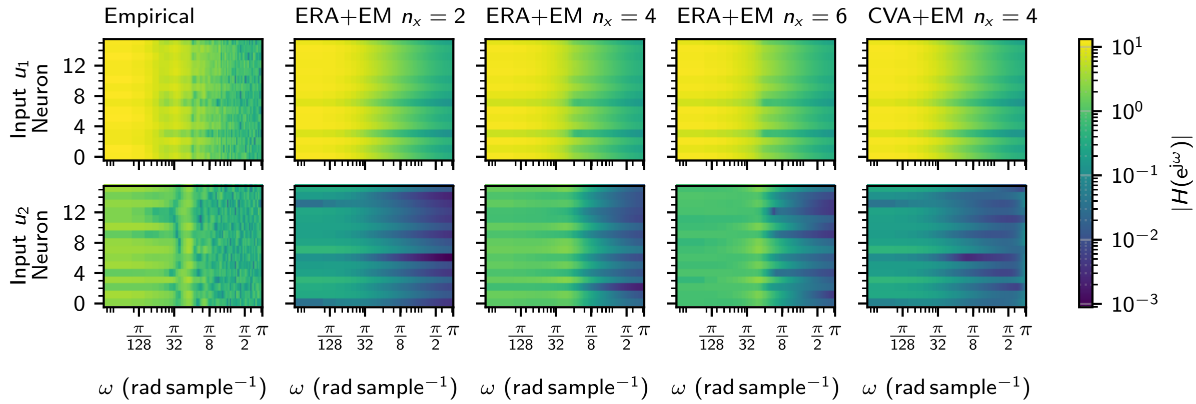

Figure 5: Per-neuron frequency selectivity for inputs u_1 (top row) and u_2 (bottom row). Each panel shows $|H(e^{j\omega})|$ for all $n_y = 16$ neurons (vertical axis) across $\omega \in [0, \pi]$ rad sample^{-1} (horizontal axis), on a shared logarithmic colour scale. Leftmost column: empirical response (honest OLS estimate from a single 5000-step recording, DFT zero-padded to 512 points). Columns 2–4: analytic ERA+EM transfer function $H(e^{j\omega}) = C(e^{j\omega}I - A)^{-1}B$ for $n_x \in \{2, 4, 6\}$ respectively, showing the latent dimension sweep. Rightmost column: CVA+EM at $n_x = 4$, for estimator comparison at the chosen order. Source

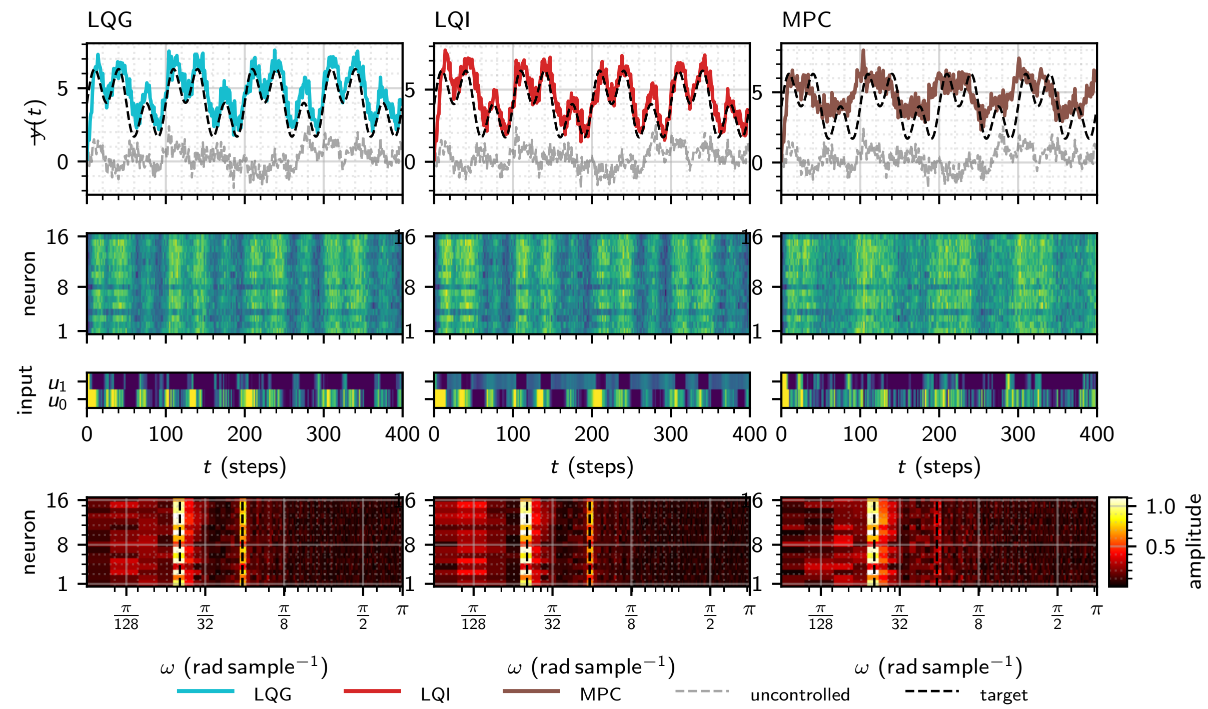

Figure 6: Closed-loop spectral control using the ERA+EM $n_x = 4$ model ($n_x = 4$), controller hyperparameters tuned by random search (8 trials each). The target is a sum of two sinusoids at $f \in \{0.01, 0.03\}$ cycles per sample with amplitudes $A \in \{1.50, 1.50\}$, centred on a physical mean of 4. Each column shows one controller (LQG, LQI, MPC) with a per-neuron tracking cost ($\lambda_{\text{var}} = 1$). Row 1: physical population mean $\bar{y}(t)$, uncontrolled baseline (grey dashed), and target reference (black dashed). Row 2: per-neuron observations $y(t)$ as a neuron $\times$ time heatmap. Row 3: applied inputs $u(t)$. Row 4: per-neuron amplitude spectral density (neuron $\times$ angular frequency ω, log scale); target frequencies shown as red dashed lines. Source